

$$1. \text{ 设 } \vec{a}=(a_1, a_2, a_3) \quad \vec{b}=(b_1, b_2, b_3) \quad \vec{c}=(c_1, c_2, c_3)$$

$$1) \vec{b} \times \vec{c} = (b_2 c_3 - b_3 c_2, b_3 c_1 - b_1 c_3, b_1 c_2 - b_2 c_1). \text{ 记 } \vec{b} \times \vec{c} = (d_1, d_2, d_3)$$

$$\vec{a} \times (\vec{b} \times \vec{c}) = (a_2 d_3 - a_3 d_2, a_3 d_1 - a_1 d_3, a_1 d_2 - a_2 d_1)$$

$$\text{左 } x \text{ 分量} = a_2(b_1 c_2 - b_2 c_1) - a_3(b_3 c_1 - b_1 c_3) = a_2 b_1 c_2 - a_2 b_2 c_1 - a_3 b_3 c_1 + a_3 b_1 c_3 = (a_2 c_2 + a_3 c_3) b_1 - (a_2 b_2 + a_3 b_3) c_1$$

$$= (\vec{a} \cdot \vec{c} - a_1 c_1) b_1 - (\vec{a} \cdot \vec{b} - a_1 b_1) c_1 = \vec{a} \cdot \vec{c} \cdot b_1 - \vec{a} \cdot \vec{b} \cdot c_1 = \text{右 } x \text{ 分量}$$

$$\text{同理, 左 } y \text{ 分量} = \vec{a} \cdot \vec{c} \cdot b_2 - \vec{a} \cdot \vec{b} \cdot c_2 = \text{右 } y \text{ 分量}$$

$$\text{左 } z \text{ 分量} = \vec{a} \cdot \vec{c} \cdot b_3 - \vec{a} \cdot \vec{b} \cdot c_3 = \text{右 } z \text{ 分量}$$

$\therefore$ 得证

$$2) \vec{a} \cdot (\vec{b} \times \vec{c}) = a_1 b_2 c_3 - a_1 b_3 c_2 + a_2 b_3 c_1 - a_2 b_1 c_3 + a_3 b_1 c_2 - a_3 b_2 c_1$$

$$\vec{a} \times \vec{b} = (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1)$$

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = a_2 b_3 c_1 - a_3 b_2 c_1 + a_3 b_1 c_2 - a_1 b_3 c_2 + a_1 b_2 c_3 - a_2 b_1 c_3 = \vec{a} \cdot (\vec{b} \times \vec{c})$$

$\therefore$ 得证

$$3) \vec{a} \cdot (\vec{a} \times \vec{b}) = \vec{a} \cdot (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1) = \cancel{a_1 a_2 b_3} - \cancel{a_1 a_3 b_2} + \cancel{a_2 a_3 b_1} - \cancel{a_1 a_2 b_3} + \cancel{a_1 a_3 b_2} - \cancel{a_2 a_3 b_1} = 0$$

$\therefore$ 得证

$$2) \begin{cases} x = 5t^3 \\ y = 4t \end{cases} \Rightarrow t = \frac{y}{4} \Rightarrow x = 5 \times \left(\frac{y}{4}\right)^3 = \frac{5y^3}{64}$$

$$\therefore 64x = 5y^3 \Rightarrow x = \frac{5y^3}{64}$$

(b) $t=0$. 位于 $(0,0)$

$t=2$. 位于 $(40,8)$

$\therefore$ 位移为 $x = (40,8)$. $\vec{v} = \frac{(40,8)}{2} = (20,4)$

$$\vec{v} = \frac{d\vec{r}}{dt} = 15t^2 \hat{i} + 4 \hat{j} \quad \text{m/s}$$

$t=0$. $\vec{v} = 4\hat{j}$ $t=2$ $\vec{v} = 60\hat{i} + 4\hat{j}$

$\therefore \vec{a} = \frac{60\hat{i} + 4\hat{j}}{2} = 30\hat{i} + 0\hat{j} = (30,0) \text{ m/s}^2$

(c) $\vec{v} = 15t^2 \hat{i} + 4\hat{j}$

$\vec{a} = 30t \hat{i}$

(d) $\hat{c} = \frac{\vec{v}}{|\vec{v}|} = \frac{15t^2 \hat{i} + 4\hat{j}}{\sqrt{225t^4 + 16}} = \frac{(15t^2, 4)}{\sqrt{225t^4 + 16}} = \frac{15t^2 \hat{i} + 4\hat{j}}{\sqrt{225t^4 + 16}}$

$(15t^2, 4) \cdot (-4, 15t^2) = 0$

$\therefore \hat{n} = \frac{-4\hat{i} + 15t^2 \hat{j}}{\sqrt{225t^4 + 16}}$

$\rho = \frac{v^2}{a_n}$ $a_n = \sqrt{a^2 - a_t^2}$

$a_t = \vec{a} \cdot \hat{c} = 30t \hat{i} \cdot \frac{\vec{v}}{|\vec{v}|} = \frac{(30t \hat{i}, 0) \cdot (15t^2 \hat{i} + 4\hat{j})}{|\vec{v}|} = \frac{450t^3}{\sqrt{225t^4 + 16}}$

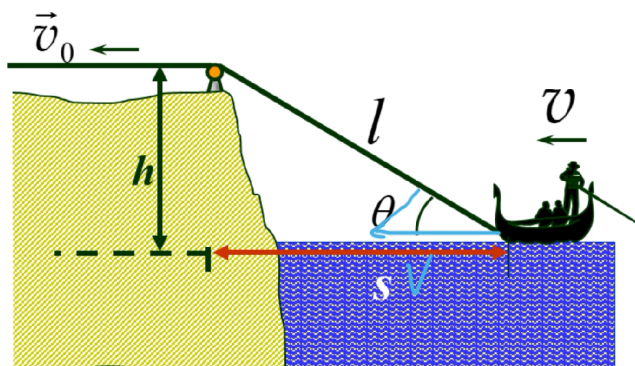
$\therefore a_n = 900t^2 - \frac{450t^3}{\sqrt{225t^4 + 16}} = \frac{(120t)^2}{\sqrt{225t^4 + 16}}$

$\therefore a_n = \frac{120t}{\sqrt{225t^4 + 16}}$

$\therefore \rho = \frac{(225t^4 + 16)^{\frac{3}{2}}}{120t}$

(e) $a_T(t) = \frac{450t^3}{\sqrt{225t^4 + 16}}$ $a_N(t) = \frac{120t}{\sqrt{225t^4 + 16}}$

3.



4. 湖岸上的人看到小船顺流而下，求小船的速度

$$\text{解: } \cos\theta \cdot v = v_0 \Rightarrow v = \frac{v_0}{\cos\theta}$$

$$l = \sqrt{h^2 + s^2}, \quad \cos\theta = \frac{s}{l} = \frac{s}{\sqrt{h^2 + s^2}}$$

$$\therefore v = \frac{\sqrt{h^2 + s^2}}{s} v_0$$

$$4. v = 110 \text{ km/h} = \frac{110 \times 10^3 \text{ m}}{3600 \text{ s}} = \frac{1100}{36} \text{ m/s} = 30.6 \text{ m/s}$$

$$y = \frac{1}{2}gt^2, \quad x = 30.6t$$

$$\text{着陆时 } \frac{y}{x} = \frac{1}{1} \Rightarrow \frac{1}{2}gt^2 = 30.6t \Rightarrow 4.9t = 30.6 \Rightarrow t = 6.245$$

$$x = 30.6t = 191.09 \Rightarrow L \approx 270 \text{ m}$$

$$5. (1) \omega = \frac{v}{R}, \text{ 径向: } \frac{dr}{dt} = v_r$$

$$\therefore \left(\frac{dr}{dt}\right)^2 + (r\omega)^2 = \left(\frac{dr}{dt}\right)^2 + \frac{v^2 r^2}{R^2} = v^2$$

$$\therefore \left(\frac{1}{v} \frac{dr}{dt}\right)^2 + \left(\frac{r}{R}\right)^2 = 1$$

$$\therefore \frac{dr}{dt} = v \sqrt{1 - \left(\frac{r}{R}\right)^2}$$

$$\therefore v_r = v \sqrt{1 - \left(\frac{r}{R}\right)^2} \Rightarrow v = v \sqrt{1 - \left(\frac{r}{R}\right)^2} \vec{e}_1 + \frac{vr}{R} \vec{e}_2, \quad \vec{e}_1 \text{ 为径向单位向量, } \vec{e}_2 \text{ 为切向单位向量}$$

$$v_0 = \frac{vr}{R}$$

$$\frac{dr}{dt} = \frac{dv_r}{dt} = v \cdot \frac{1}{2\sqrt{1-\frac{v^2}{R^2}}} \left(-\frac{2v}{R^2}\right) \frac{dr}{dt} = -\frac{v^2}{R^2} r$$

$$a_r = -\frac{v^2 r}{R^2} - \frac{v^2 R}{R^2} = -\frac{2v^2 r}{R^2}$$

$$V_{t\eta} = \omega \cdot r \quad a_{\theta} = \frac{dv_{t\eta}}{dt} = \frac{d(\omega r)}{dt} = \omega \frac{dr}{dt} = \frac{v^2}{R} \sqrt{1-\left(\frac{r}{R}\right)^2}$$

$$\therefore a = -\frac{2v^2 r}{R^2} \vec{e}_1 + \frac{v^2}{R} \sqrt{1-\left(\frac{r}{R}\right)^2} \vec{e}_2$$

$$(4) \frac{dr}{dt} = v \sqrt{1-\frac{r^2}{R^2}}$$

$$\frac{d\theta}{dt} = \frac{v}{R}$$

$$\Rightarrow \frac{dr}{d\theta} = R \sqrt{1-\frac{r^2}{R^2}} = \sqrt{R^2 - r^2}$$

$$\therefore \frac{dr}{\sqrt{R^2 - r^2}} = d\theta$$

$$\therefore \arcsin \frac{r}{R} = \theta$$

$$\therefore r = R \sin \theta$$

$$b. V_{\text{相对}} - V_k = \frac{2}{1} = 2 \text{ km/h}$$

$$(V_{\text{相对}} + V_k) t_{\text{合}} = 5 \text{ km}$$

$$\text{瓶子向下运动 } 3 \text{ km, 时间 } t = 1 + t_{\text{合}}$$

$$\therefore V_k (1 + t_{\text{合}}) = 3$$

$$V_{\text{相对}} = V, \quad V_k = V_v, \quad t_{\text{合}} = t_1$$

$$\begin{cases} V_1 - V_2 = 2 \\ (V_1 + V_2) t_1 = 5 \\ V_2 (1 + t_1) = 3 \end{cases}$$

$$V_2 + V_2 t_1 = 3 \Rightarrow \frac{3 - V_2}{V_2} = t_1$$

$$\begin{cases} \frac{V_1 + V_2}{V_2} (3 - V_2) = 5 \\ V_1 = 2 + V_2 \end{cases}$$

$$\frac{3V_1 + 3V_2}{V_2} = 5 + V_1 + V_2$$

$$\frac{6 + 3V_2 + 3V_2}{V_2} = 5 + 2V_2 + 2$$

$$6 + 6V_2 = 7V_2 + 2V_2$$

$$\therefore 2V_2 + V_2 - 6 = 0 \Rightarrow x = \frac{3}{2} \text{ km/h}$$

$$\therefore V_* = \frac{5}{2} \text{ km/h}$$

$$V_{\text{rel}} = \frac{7}{2} \text{ km/h}$$